

Institute of Actuaries of India

Subject SA7-Investment and Finance

June 2019 Examination

INDICATIVE SOLUTION

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i) The key areas that would merit consideration are

1. Board/Management quality – various factors to be considered here
 - a. Do they have industry experts and technology experts with them? [½]
 - b. How much of Indian property market experience do they possess? How well do they understand the dynamics and buyer/seller sentiments for the primary property segment (where under-construction properties are sold)? [1]
 - c. Does the Board/Management possess similar expertise in other foreign markets? Since it is a foreign PE fund does it have similar investments in online property broking. [½]
 - d. What is the management's view on the compliance issues plaguing the property industry? Basis discussions with the management as part of the process of valuing the Company this can be gauged to a reasonable extent. [½]
 - e. Track record of key executives within the Company and Board. [½]
2. Market Growth Prospects [3]
 - a. Urban, Tier 1 and Tier 2 cities in India has a growing middle class which is also quite computer savvy. Hence the demography is very supportive. [½]
 - b. Home ownership is a traditional value instilled in most middle/upper middle-class segments of the Indian population - so demand for the business the Company operates in is likely to be high. [½]
 - c. More and more players entering tech-enabled business – so the traditional model of connecting buyers/sellers is losing appeal amongst the younger generation who are earning well and are keen to buy their property as early as possible. [½]
 - d. Given that property has been a growing asset class in India – especially the past few decades having witness substantial growth – the industry would continue to be attractive and the competition is expected to be fierce. [1]
 - e. New regulations like RERA (Real Estate Regulation Act, 2016) have provided further security and comfort to buyers – this in turn increases confidence of the buyers leading to higher demand. Especially about the working-class home buyers, RERA has provided a long sought-after relief. [½]
3. Competition [3.5]
 - a. Likely aggressive competition given that this is a growing space.
 - b. Moreover, it is technology driven – and hence the competition is expected to be sophisticated. Those with deeper pockets are likely to compete hard.
 - c. Regulations like RERA that has been recently launched has also instilled a great amount of discipline in the competition – cutting off unhealthy practices that were not in the best interests of the consumers or the industry.
 - d. Since brokers also require compulsory registration under RERA, and their activities can be monitored/scrutinized, rogue and unscrupulous players would anyways be on the decline.
 - e. Those with the best management team with the right skillsets across the value chain of the sales process would likely perform better. [3]
4. Product/Service Quality:
 - a. What is the process adopted by this portal to secure the transaction? A good property brokerage firm would typically play a role in each of the below areas:

- i. Matchmaking/Listing - between the available properties and the needs of the registered buyer. Various factors such as location, size, budget, developer credentials, neighbourhood etc. would have to be matched. Depending upon how rich the data analytics capabilities are – the better that the portal would manage to secure a proper match. This implies, inter-alia, the systems and processes to enable such vast matchmaking to happen. So, the questions that needs to be answered are how well the portal manages to match the needs of potential buyer's basis parameters such as the ones stated above. Are there any user feedback ratings that can also indicate how good their service is? Have they been consistent in providing good service standards across time?
- ii. Fulfilment of the potential property – these would include site visits to ensure that whatever was depicted online is a fair representation of reality. Does the Company have specialized staff who can assist in the site visits?
- iii. Loan capacity evaluation - Once the property/properties have been shortlisted after site visits– the loan capacity evaluation also happens. The que
- iv. Negotiation with the property developer builder – best prices, best location within the project.
- v. Legal formalities including assistance in paperwork completion. [3]
- b. Do they have informative articles and research to help potential buyers understand the market dynamics and help them make an informed decision? [1]

5. Input Costs:

- a. These are primarily technology driven. The main costs are therefore the development costs involved in this highly competitive e-commerce space.
- b. Technology costs comprise of both fixed and variable – arguably the former would be the heavier component. The fixed costs in turn would get averaged out across the various sales transactions – hence depending upon the average number of sales transactions concluded through ghar.com (including any projected growth in the same) the break-even point would get influenced.'
- c. Also, amongst the employees involved on the technology side might have a few highly paid resources. These are primarily costs involved in the "matchmaking" aspects and the "Sale conclusion "aspects. [2]

6. Retained Profits

- a. Likely not very high since these companied have heavy overheads and longer break-even terms.
- b. However, the size of the parent/investor pockets can be a suitable proxy to gauge the level of risk appetite and potential market aggression. [1]

7. Risks

- a. What if projects get delayed due to issues being faced by reputed builders. How would the payment stream to X Ltd. be affected?
- b. Delay in construction can cause delays in tranche payments and hence delays in revenue realizations for ghar.com.
- c. Compliance with regulations can further impact delivery – impacting revenue flows of X Ltd.
- d. Some of the builders may face severe liquidity issues and shut down operations temporarily.
- e. Also, with recent government interventions like demonetization of high value currency notes, property prices crashed dramatically leading to massive cancellations of

bookings made by buyers (since the expected market value of the property has fallen significantly and the mortgage value to property value ratio is close to 100%). [4]

8. History

- a. The track record of the company measured in terms of customer satisfaction would play a paramount role.
- b. User reviews on the various aspects of the sale process – both matchmaking and transaction closure – would help place a value on the brand ghar.com
- c. Also, how they have handled any potential crisis (like demonetizations etc.) would become equally relevant.

[25]

ii) 3 parameters

Parameter	Definition
Group Ego	The difficulty of moving a narrative away from a popular or accepted norm, because others in the group will have their egos hurt, and others in the group will want to avoid hurting egos.
Conforming	the tendency for people to confirm an accepted narrative. This can be enhanced by education, which normally teaches and established method for analysis.
Herd like behaviour	The tendency from groups to find comfort in adopting a common belief or a common approach it revolves around the idea of strength in numbers, and the benefit that it gives to the egos of those in the group.

Alternatively, marks can also be awarded to

Nous : as 'mind', 'thought', 'forethought', 'imagination', 'intuition', 'reflection' and the 'mind's eye' amongst others. We will use it to mean the way in which the individual views or perceives the world.

Logos is translated as 'logic' or 'rational thinking'. It refers to the type of rational logic that could be understood by all. The function of Logos is to condition the mental picture created by Nous.

Pyr is translated as 'fire', 'fire in the belly', 'animal spirits', or 'impulse'. It refers to the light and energy used by the Psyche.

[3]

iii)

Parameter	Buyers	Sellers
Group Ego	- For both investor and middle class - invest in top builders who have large projects (pay a premium). This should mostly be beneficial. [1/2] - For middle class – buy properties in larger societies with amenities (rather than builder floors). These societies might not provide the best value for money sometimes – depending upon that it may or may not be beneficial. [1/2] - Given the strong belief that it is okay to buy a home even via mortgage – it could become compulsive to buy a home on	- Given that middle class would rather invest in homes by leveraging (mortgages) this Group Ego narrative of buyers would likely benefit the sellers a lot. [1/2] - Use potentially lower quality plumbing, flooring etc. since the buyers are more concerned on location/neighbourhood etc. and the other things they will anyways enhance it themselves.

	mortgage even though it might be financially the best decision. [1]	This narrative could potentially damage the reputation of the builder on the long run. However, if all the participants behave similarly -then it results in a disadvantage for the market. [1/2]
Conforming	- Home ownership is mandatory – you must strive to buy your own home – (narrative mainly -for middle class). [1/2] - Houses are a -good asset class so invest in them given long term returns (mainly investor class) [1/2] - This could potentially be good for the buyer – if it is not very leveraged (high proportion of mortgage). Also, there could be additional cashflow advantages (including tax advantages) of owning a property and renting it out whilst staying at another property. [1/2] - Depending upon whether the property is bought as an investment or self-occupation – the advantages of conforming can differ. For self-occupation there is a clear advantage in conforming. For investor class – depending upon future property price movement – they may or may not be successful. [1] - Alternatively, a land investment could be more beneficial than property investment but given the “Conforming aspects” [1/2]	- Attract buyers with celebrity advertising. These could prove costly and if there are any subsequent project losses – then the advertising cost would be dear! [1/2]
Herd like behaviour/Market Psychology	- Property prices always are on the rise. [1/2] - Potential Buyers would not defer consumption - might simply take a loan to fulfil their consumption. [1/2] - Depending upon how the loan to home value ratio moves – this could be either detrimental or beneficial to the buyer. [1/2]	- The herd-like behaviour of buyers would only benefit the sellers by harping on the same. [1/2]

Alternative solution of using Nous, Logos and Pyr is also be acceptable.

iv)

- a) EMH – Asset prices fully reflect all available information. The consequence is that the property stocks always trade at their fair value. This means it would be impossible to beat the market consistently by picking individual stocks.

If EMH were to hold to a large degree – then people would not be willing to pay for research since there would be no consistent or significant value in doing so. EMH would support passive management overall as opposed to active management.

The number of trades and the cost of trading would consequently be lower if EMH were to hold since investors cannot exploit outperform consistently over a long period in an active manner. Trading volumes would remain low, therefore. [3]

- b) Deficiencies in EMH narrative (book-work – point 4.2 of chapter 10 of SA 7).
- a. Asset Prices do not reflect all available information
 - i. Amount of information enormous. Volume and complexity too much even for a competitive self-aware investor.
 - ii. Scarcity of time and resources to take account of all market inputs.
 - iii. Consequently, limits placed on narratives.
 - iv. Dominance of herd-like behaviour likely limits amount of information fully incorporated into market prices.
 - v. EMH argues that only a few well-informed investors are needed to bring market price to fair value. In practice this has not been the case.
 - b. Investors are not rational
 - i. Rationalizations basis other's narratives
 - ii. These are basis collective thinking resulting in herd-like behaviour.
 - iii. Rationalizing someone else's narrative is not the same as being rational.
 - iv. Market participants are not fully engaging their Psyche (Nous and Logos mainly)
 - c. EMH is a Group Ego defense mechanism
 - i. Various anomalies have been shown to exist – these invalidate EMH.
 - ii. However, group ego defense mechanisms play it down since the anomalies are analyzed within the narrative of EMH.
 - iii. One can argue that EMH is a justification of average investment performance – sought by those who feel good sticking to a poor-quality narrative.
 - iv. Lot of psychological feel good benefits for narratives played out over a long period. The Group Ego defense mechanism is a way of protecting the un-competitive or lazy investor to explain his performance.
 - d. Investors with market power
 - i. EMH ignores players with market power. These people can significantly influence the outcomes of the price.
 - ii. They can effectively bully other investors.
 - iii. Unlevel playing field since few players have disproportionately higher power than others. This makes it unlikely that the prices would trade at fair value.16

[8]

c) Possible Framework

Likely Successful	Likely Unsuccessful – early warnings for course correction
Understands the implication of opposing theories. example EMH etc.	does not properly understand rival theories
exercises patience and humility	is mostly impatient and overconfident
considers it essential to have a disciplined psychological approach to investing	wants to try to get rich quickly and without discipline
actively and critically looks at the world and creates a critical narrative off it in a sceptical manner.	Creates a cynical narrative of the world
create deep narratives talking when looking for causes and effects	creates shallow narratives when looking for causes and effects
sees a necessity to dig deeper behind media narratives; is sceptical of face values	Missy news on face value or in the context of conspiracy theories
when creating their own narrative want this to be as realistic as possible; questions existing narratives in a progressive manner to improve understanding	when creating their own narrative, may suffer from distortions, dismisses existing narratives in an overly negative way
are disciplined with their losses	have little discipline with their losses
continue to learn to improve their narratives	continue to add unjustified or unverified ideas to their narratives
ability to persist and start over again try to learn from their mistakes	ability to persist and start over again but they usually don't learn from their mistakes
understand the advantages and disadvantages of diversification	does not have a good understanding of probability and statistics

[8]

[55 Marks]

Solution 2:

i)

There are three key banking risks

Credit risk - the risk that loans are not repaid

Liquidity risk- the risk that customers do not act independently but panic and want their money back at the same time

Margin risk- the risk that the rate paid for short-term deposits varies while the rate charged on long-term lending remains fixed. [3]

Managing credit risk

This is a two-stage process. First banks need to recognize that some loans will not be repaid. This is done by generating credit scoring models that predict whether an individual loan will be repaid based on experience of similar loans. i.e. underwrite the loan.

Past claims are recorded along with customer characteristics to see which characteristics correlate to higher credit losses. Now a days there are data models which helps predict the customer delinquency.

The second stage to manage credit risk is recognizing that credit losses fluctuate significantly through time. While banks can try and predict an average rate, recessions, increases in unemployment or collapses in house prices can trigger much larger losses whose severity and timing are unknown. [2]

Managing liquidity risk

Banks could remove liquidity risk by matching the duration of assets and liabilities, but this would also remove their main source of profit from borrowing short and lending long.

Managing this risk is difficult as it is effectively related to confidence. If customers are confident the bank is sound, they are unlikely to all want their money back at the same time, but if they are worried, they demand their deposits back at the same time and the subsequent bank run will destroy any bank.

Banks can ensure they are so well managed that there is minimal risk of a loss of confidence. Part of this would involve holding (much) higher levels of capital. Another part would be holding excess liquidity. Banks can also hold a so-called 'liquidity buffer'. This is a pool of highly liquid assets that can be sold quickly to realise cash in case of a bank run. [2]

Managing margin risk

Banks borrow short-term money and offer savers the prevalent market interest rates (often related to the central bank's base rate). However, they lend out longer term, perhaps in the form of fixed-rate mortgages that make it hard to re-price the rates for borrowing customers to match changes in the rates paid for deposits.

This temporal variation creates a significant risk for banks. Banks can mitigate this risk to a certain extent by undertaking 'pay fixed, receive floating' swaps. If floating rates rise, and the interest rate on bank deposits rises, then the swap will become valuable. This should offset the fact that the mortgage asset, which has a fixed rate, does not reprice to the higher interest rates whereas the bank deposit cost will rise for the bank. [2]

Note: Solutions covering Operational, Market, Maturity transformation risks etc. are also acceptable.

[9]

ii)

Tiers of capital

The following table describes the forms of capital and where each would be categorised:

Tier 1	Share Capital (equity)
	Disclosed reserves
Tier2	Hidden reserves
	Unrealised gains on investment securities Medium to long-term subordinated debt
Tier 3	Short-term subordinated debt

[3]

iii)

a. Limit on cash withdrawal

Such a limit would save the five banks from a liquidity crisis, in that they should be able to provide the amount of cash needed on a daily basis.

However, investors and depositors would be queuing every day in front of cash machines to withdraw the maximum cash amount. This has a negative impact on confidence and can lose many hours of productivity over a very long period.

The crisis may not resolve itself in the short term, and the limit may stay in place for years.

Depositors in other banks would be penalised even if their banks are safe.

The economy depends on cash for certain activities (*e.g.* small-scale shopping, restaurant and drinks industry, certain building trades). These may see a knock-on crisis if cash is limited.

If the limit avoids central bank loans, it will avoid the taxpayer being put at risk in the crisis.

Depositors will think hard about where to place their cash in future, so moral hazard is reduced.

If the banks have overseas subsidiaries it would not be possible to impose the cash ban in the other countries. This would allow overseas depositors to get out of the banks, when domestic depositors cannot. [4]

b. Act as lender of last resort

This would involve giving unsecured loans to the banks that are in danger of collapse. It has the benefit that it would provide liquidity for the banks to continue trading and provide cash for those customers that wish to withdraw capital.

Depending on the sizes of the banks, the amount of lending required could be extremely large.

Any central bank bail-out produces a moral hazard whereby the investing public believe that there is no risk to their capital, and do not make the risk/return assessment required before placing their funds with a bank. This favours the bank that offers the highest rate on deposits, even if it is a risky enterprise. If the public learn that such bailouts are occurring, a run on the banks concerned would be inevitable as investors move money to solvent banks 'just in case'.

This would make it impossible to support the five banks indefinitely.

It avoids a domino effect, whereby lack of confidence in the three struggling banks undermines confidence in the whole banking system. This could produce a run on all commercial banks.

The collapse of the banks could cause severe recession in the general economy, leading to the collapse of the other banks and the collapse of many non-banking entities. Lender of last resort could avoid this disaster scenario.

Ultimately the taxpayer is paying for the rescue of some investors who chose the wrong investment. This is unfair to the investors who chose the right investment and undermines the capitalist system of economics. [4]

c. Allow collapse

The taxpayer is insulated from the bad decisions of the depositors in the failing banks.

So long as the five banks do not represent a significant part of the banking sector, the other banks should in time be able to buy the remaining business from the failing banks at reasonable prices.

Depositors in the failing banks are very unlikely to lose all their deposits.

The impact on confidence could spark a larger crisis in the economy and for the remaining banks.

The central bank is not placed at risk, so ultimately it should be able to support the wider economy.

Under the other options, the risk that the central bank goes bust is a genuine one, which would undermine the currency itself.

If banks make losses, they burn up capital resources. They could then breach international Basel banking standards and be banned from certain other international economies, which would further increase the chances of total collapse. [4]

[12]

iv)

- a. Three market participants who enable markets to operate
Dealers, market makers, traders, exchanges, clearing houses, custodians
Three who offer services
Investment analysts, fund managers, strategists, consultants, actuarial consultant, advisors.

[3]

b. The merits in friend's view

Better models may have captured some of the risks that led to the banks going bust, which were perhaps missed by the existing models.

Capital levels were based on the results of existing models, and capital buffers may have been higher if the models had more accurately predicted the future risks.

If all banks used a variety of different models, then perhaps some models may have correctly predicted the risks and others may not have predicted the risks, which is better than all banks using a similar model to one another and therefore all being exposed to the same uncaptured risks.

Criticisms of the Friends view

The models were presumably the best that the banks could create at the time.

The only data that is available to calibrate the models would have been the past data, and the events that caused the collapse may not have occurred in the past.

Overcomplicating models can have a negative effect whereby users assign a degree of confidence to the results that is not warranted, and the model can become unstable due to the detail and complexity involved.

[4]

c. The merits of your views

Financial markets are driven by real world events, which are influenced by political decisions at home and abroad, as well as economic and natural shocks that may originate at home or abroad. It is arguably impossible for any quantitative measure to capture these human or unpredictable events.

Markets move in cycles, and predicting the stage of the cycle, and when the cycle may end is a matter of human judgement, which cannot be predicted by computer models.

Criticisms of your view

In most market circumstances, the past is a reasonable predictor of the future, therefore models serve a purpose, and the better the model can be designed, the better information the bank will receive.

Moving the emphasis to subjective and qualitative measures would introduce a lack of uniformity into the banking system that would have a negative impact on competition, level playing fields. It may significantly increase the risk of bankruptcy.

[4]

[35 Marks]**Solution 3:****i) The theory being referred to is Dividend irrelevance theory**

This theory holds that -in a perfect world without taxes, transaction costs or other market imperfections - the market valuation of a company is unaffected by dividend policy. It is argued that, since an investor's desire for additional cash could be satisfied by selling part of an existing investment holding, investors will not pay a higher price for shares of firms with high payouts.

Equally, if a firm wished to increase its dividend payout but wished to alter neither its investment nor its borrowing policy, then the extra dividends could only be paid if, at the same time, new shares were issued and sold to new investors. In a perfect market, those investors will only be willing to pay a price for shares that equalled their value. Since all real aspects of the firm are unchanged, the value of the new shares issued must equal the value of the dividend paid.

Assumption underlying the theory is Market is efficient and investors act rationally

Several practical factors are influential in setting dividend policy:

- Target dividend payout ratios.

A company may have an aim not to deviate too far from a certain level of payout. The level will vary dependent upon factors such as the nature of the company's business, its age and size.

- Trends in long-run, sustainable earnings.

The company will wish to target a stable dividend policy that can be supported by its typical level of operations. This means it is less likely to take account of short-term cashflow changes or extraordinary profits in setting its regular dividends (although short-term positive cashflows may be reflected in one-off special dividend payments).

- Administrative and processing costs.

These can form a significant portion of the cost for small dividend payments. This may mean that it is not sensible to distribute a dividend.

- Impact of *changes* in dividends.

The company will consider the relative size of the dividend payment from one payment to the next - big changes will lead to market comment.

- Wish to avoid future reductions in dividends.

These are generally viewed unfavourably by the market.

- The information content of dividend policy.

In countries where investors have little confidence in other sources of company information (*e.g.* the quality of the accounts), great store will be placed on the dividends as an indication of the company's true financial position. [5]

ii)

- Company's profits have substantially decreased, and directors are not confident about the future trend of profits. The current levels may not be sustainable.
- Some internal business / product failures which has made the directors cautious and hence they prefer to set a lower expectation.
- There may be a market expectation that dividends may fall. This could be due to various reasons and directors may want to maintain the expectations.
- It may be that the Pharma industry and hence competition has reduced as well. Directors may want to use the opportunity to revise.
- Cash reserves may have dipped.
- It could be due to a long-term strategy to reduce dividends and use the cash for other investments.
- There could be taxation changes which means it is no longer attractive to pay-out dividends
- Any statutory restrictions which constraints the company to pay out. [5]

[10 Marks]
